

Learning Objectives: SCRUM

After completing this chapter, students will be able to:

1. Understand the fundamental concept and origin of SCRUM

Explain the purpose, historical background, and motivation behind SCRUM as an agile software development framework, including its emphasis on adaptive, fast, and self-organizing development teams.

2. Analyze the SCRUM process model and its major phases

Describe the complete SCRUM lifecycle and differentiate among the three major phases:

- Pre-game Phase
- Development (Game) Phase
- Post-game Phase

3. Explain the activities performed during the Pre-game phase

Identify how planning and architecture activities are performed before development begins, including requirement identification, Product Backlog creation, priority assignment, effort estimation, and system architecture planning.

4. Evaluate the role of Sprint-based development in SCRUM

Describe how Sprints support iterative software development by producing working product increments through repeated cycles of requirement analysis, design, implementation, and delivery.

5. Differentiate between Product Backlog and Sprint Backlog

Compare the purpose, characteristics, ownership, and usage of Product Backlog and Sprint Backlog in managing software requirements and development activities.

6. Analyze SCRUM roles and responsibilities

Explain the responsibilities and contributions of different SCRUM participants including:

- Scrum Master
- Product Owner
- Scrum Team
- Customer
- Management

7. Understand SCRUM meetings and their importance

Describe the objectives and activities of important SCRUM meetings:

- Sprint Planning Meeting
- Daily Scrum Meeting
- Sprint Review Meeting

8. Apply SCRUM concepts in real-world software development scenarios

Identify appropriate SCRUM phases, roles, artifacts, and practices from practical project situations and justify their application.

9. Evaluate the importance of teamwork and self-organization in SCRUM

Explain how collaboration, team decision-making, communication, and responsibility sharing help achieve Sprint goals successfully.

10. Analyze the final release activities in the Post-game phase

Explain how integration, system testing, documentation, and release preparation activities ensure successful completion of a SCRUM project.

Important Points: SCRUM

1. Definition of SCRUM

- SCRUM is an agile software development framework.
- It supports adaptive, quick, and self-organizing product development.
- SCRUM focuses on teamwork, flexibility, and iterative improvement.

2. SCRUM Process Model

SCRUM process consists of three major phases:

Pre-game → Game Phase → Post-game

- (a) Pre-game Phase
- (b) Development (Game) Phase
- (c) Post-game Phase

3. Pre-game Phase

The Pre-game phase contains two sub-phases:

Planning + Architecture

Planning Activities:

- Define the system requirements.
- Create Product Backlog.
- Prioritize requirements.
- Estimate development effort.
- Identify team, tools, resources, and risks.

Architecture Activities:

- Prepare high-level system design.
- Plan architecture based on Product Backlog.
- Identify possible implementation problems.
- Perform design review meeting.

4. Development (Game) Phase

- Main development happens in this phase.
- Development is performed using Sprints.
- A Sprint produces a working product increment.
- Each Sprint includes:

Requirement → Analysis → Design → Development → Delivery

Sprint Duration:

1 – 4 Weeks

5. Post-game Phase

Occurs when:

- All requirements are completed.
- No new backlog items exist.
- Product is ready for release.

Activities:

- Integration
- System Testing
- Documentation
- Release Preparation

6. SCRUM Roles

Scrum Master + Product Owner + Scrum Team + Customer + Management

Scrum Master

- Ensures SCRUM rules are followed.
- Supports project progress.
- Communicates with team, customer, and management.

Product Owner

- Manages Product Backlog.
- Controls requirements.
- Makes final backlog decisions.

Scrum Team

- Self-organized development team.
- Creates Sprint Backlog.
- Estimates development effort.

7. SCRUM Artifacts

Product Backlog

- Complete list of requirements.
- Updated continuously.
- Contains priority and effort estimation.

Sprint Backlog

- Selected Product Backlog items.
- Work planned for current Sprint.
- Stable until Sprint completion.

8. SCRUM Meetings

Sprint Planning Meeting

- Organized before Sprint starts.
- Selects goals and functionality.
- Plans implementation strategy.

Daily Scrum Meeting

- Tracks daily progress.
- Discusses completed work.
- Identifies problems.

Sprint Review Meeting

- Conducted after Sprint completion.
- Demonstrates working product increment.
- Collects feedback.

9. Important Exam Memory Tricks

- SCRUM Phases:

PGP = Pre Game, Game, Post Game

- Pre-game:

PA = Planning + Architecture

- Roles:

SPTCM = ScrumMaster + ProductOwner + Team + Customer + Management

- Backlog Difference:

Product Backlog = Complete Requirement List

Sprint Backlog = Current Sprint Work